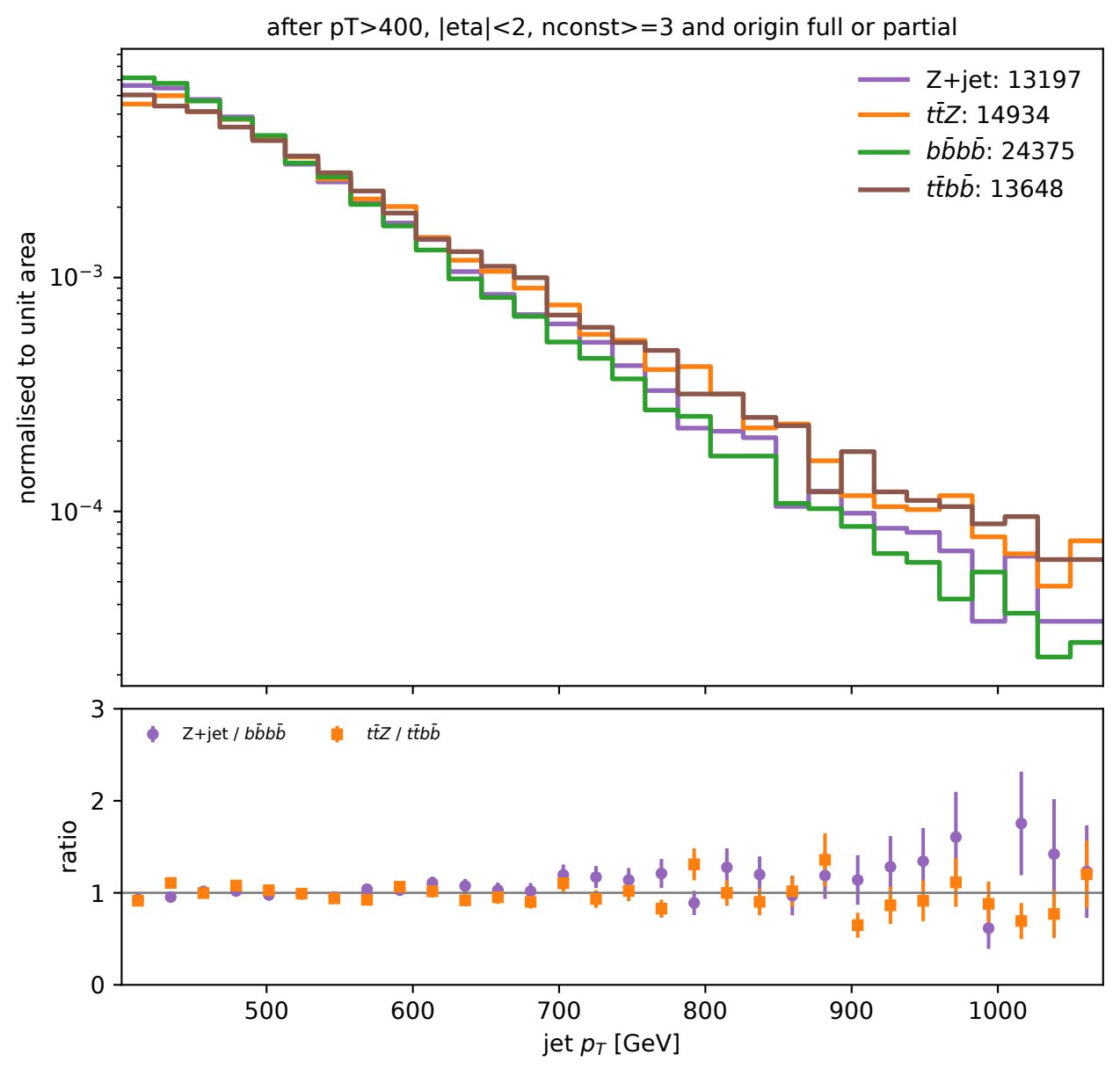
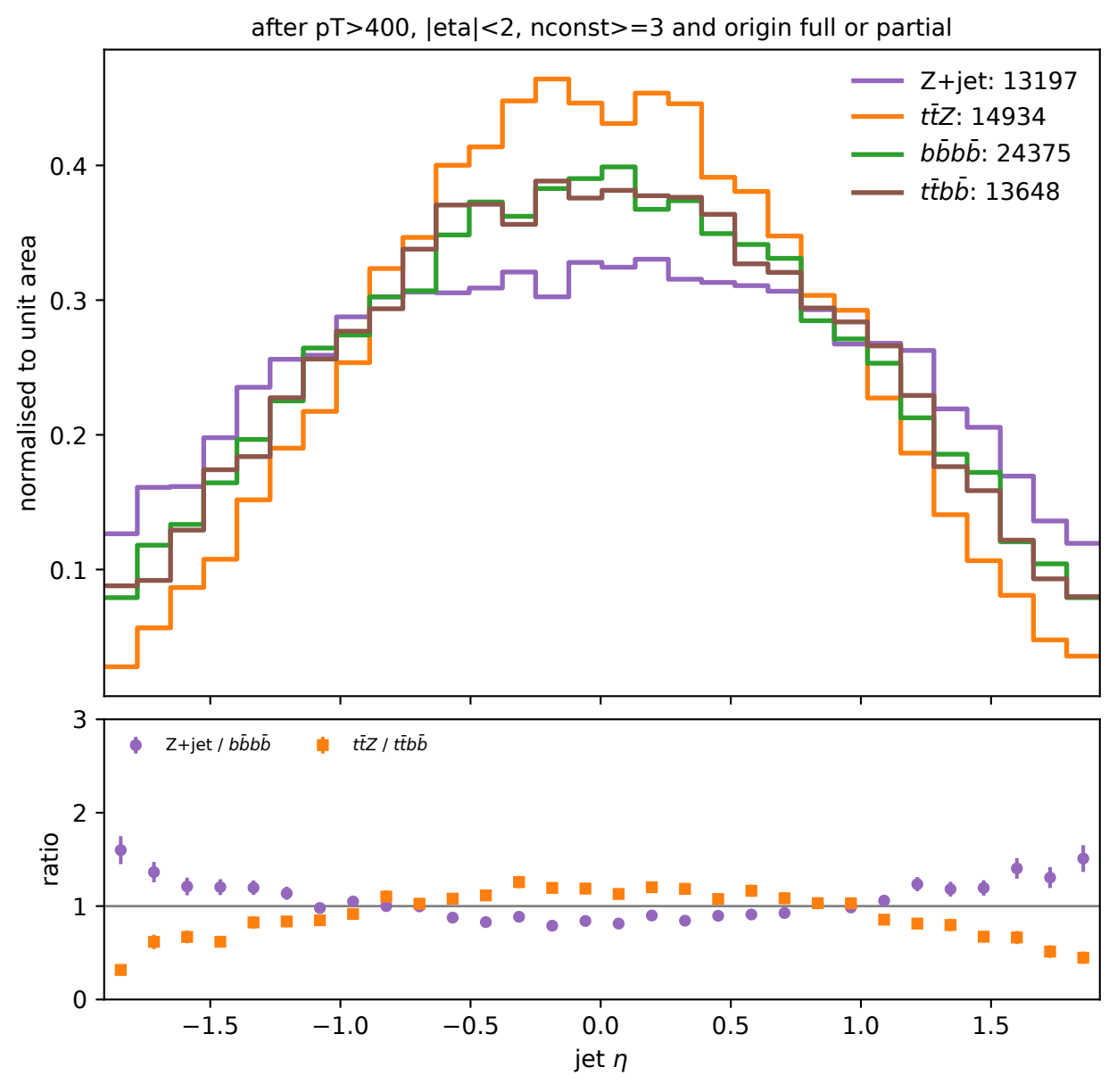
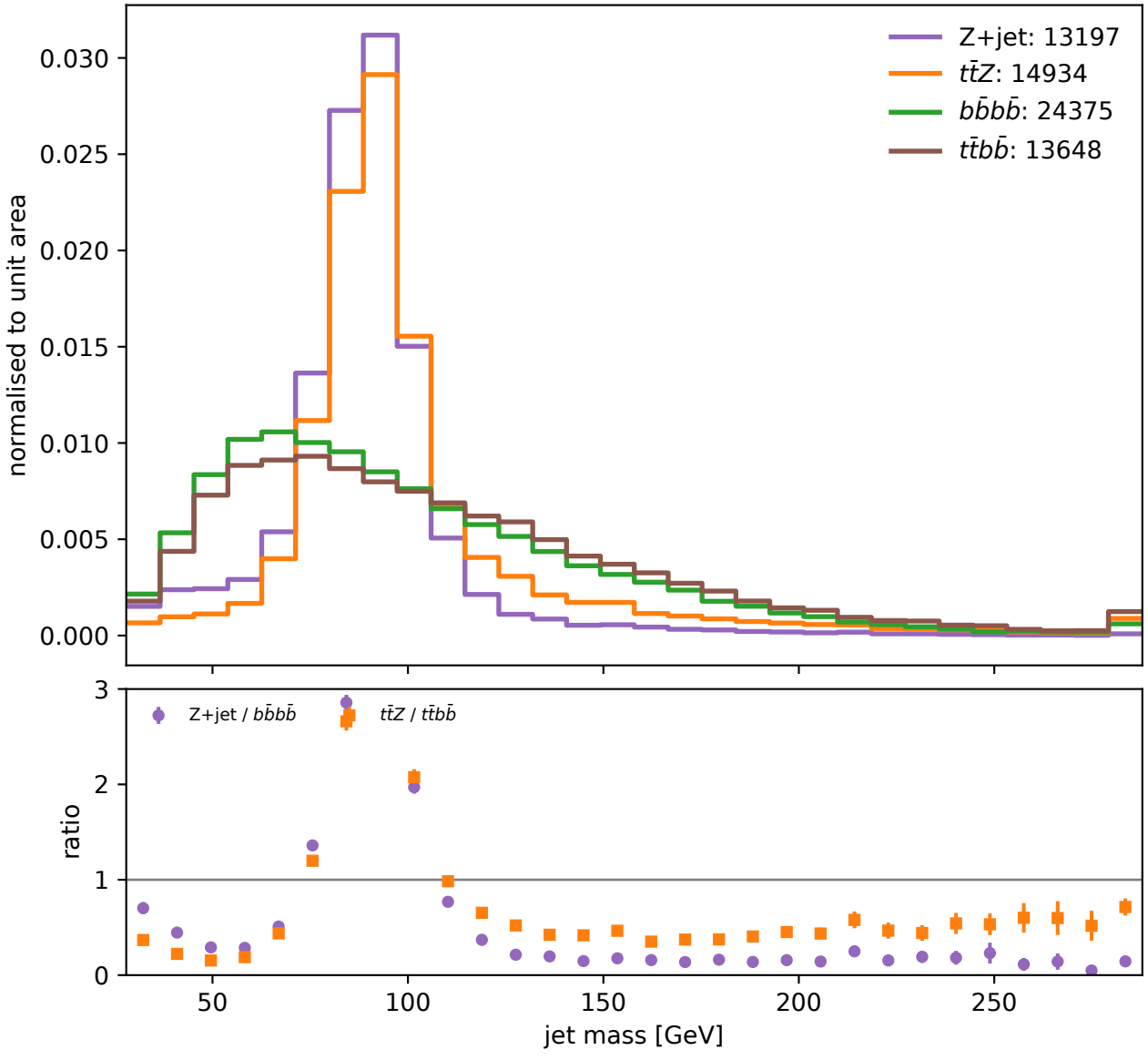
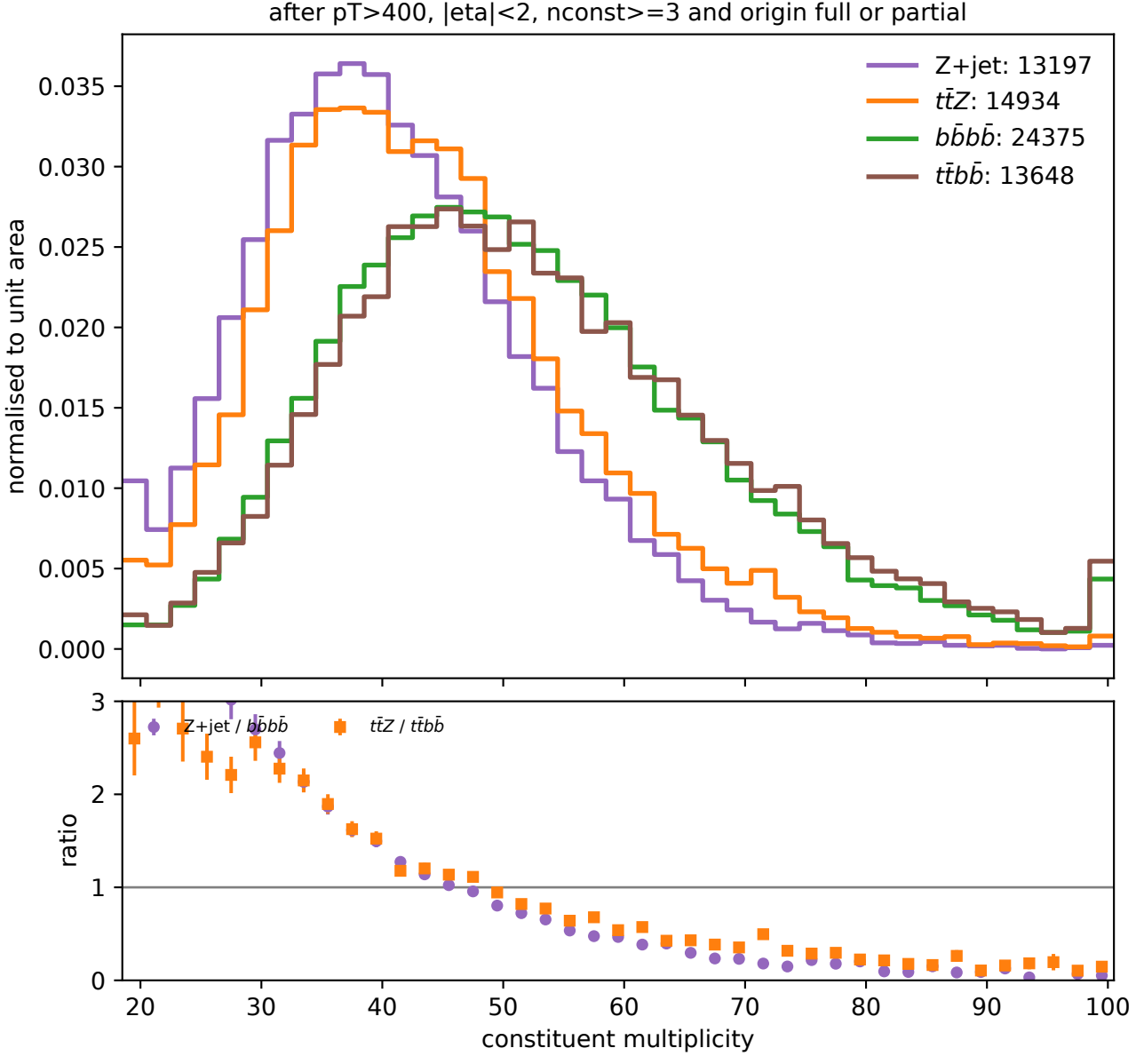
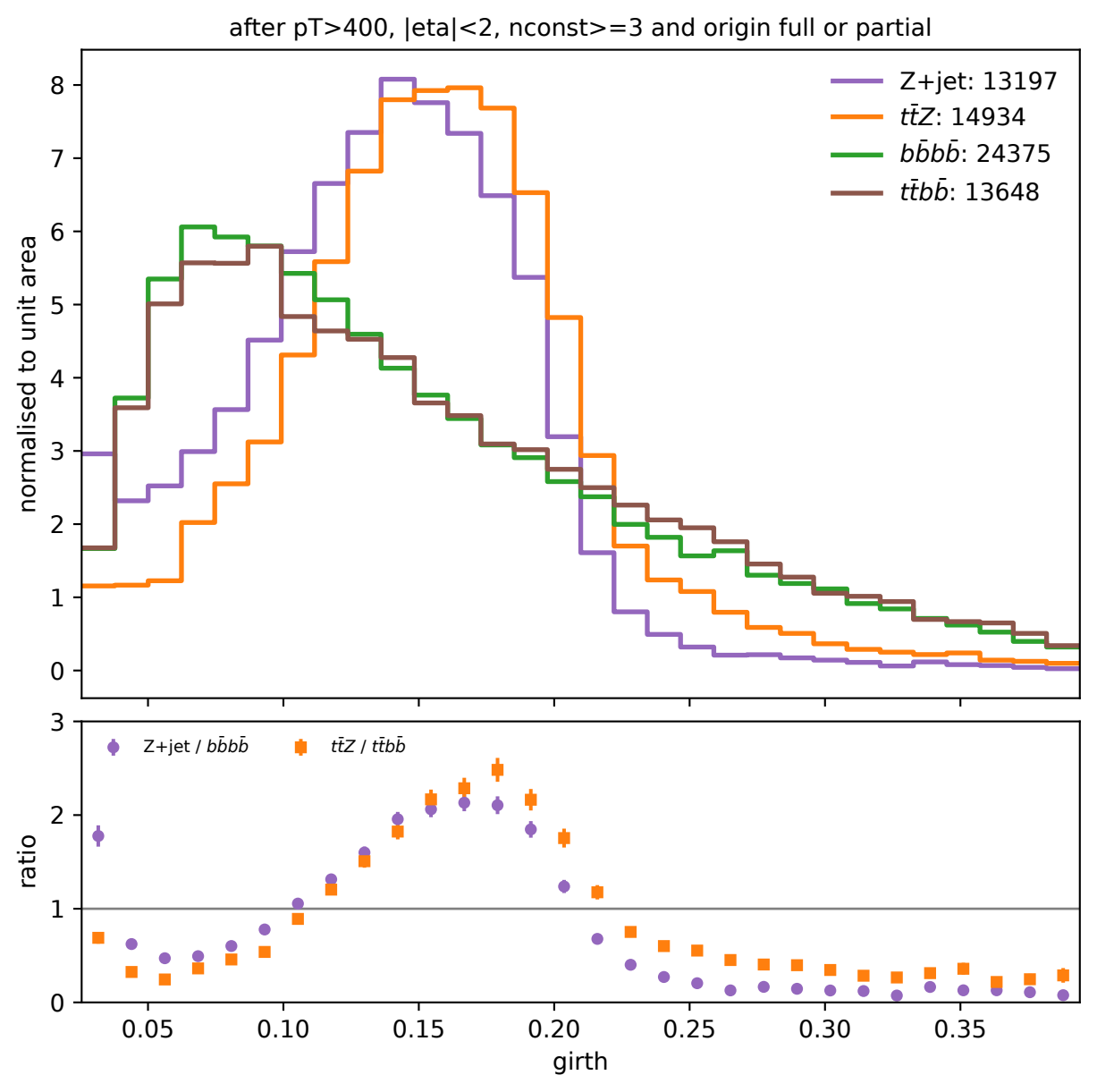
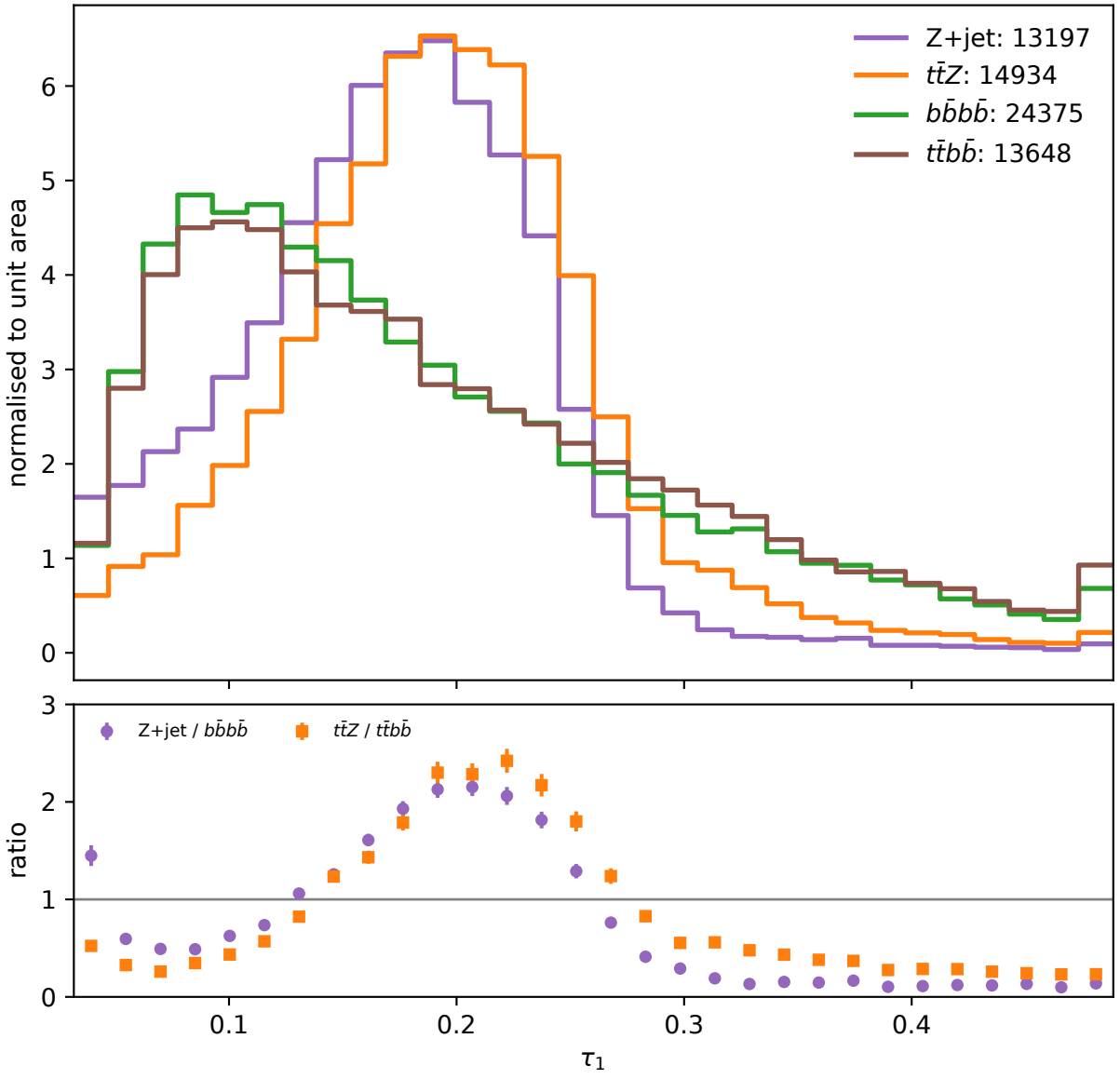

after pT>400, |eta|<2, nconst>=3 and origin full or partial
Z+jet: 13197
$t\bar{t}Z$: 14934
$b\bar{b}b\bar{b}$: 24375
$t\bar{t}b\bar{b}$: 14648
normalised to unit area
10^{-3}
10^{-4}
Z+jet / $b\bar{b}b\bar{b}$
$t\bar{t}Z$ / $t\bar{t}b\bar{b}$
ratio
0
1
2
3
500
600
700
800
900
1000
jet p_T [GeV]

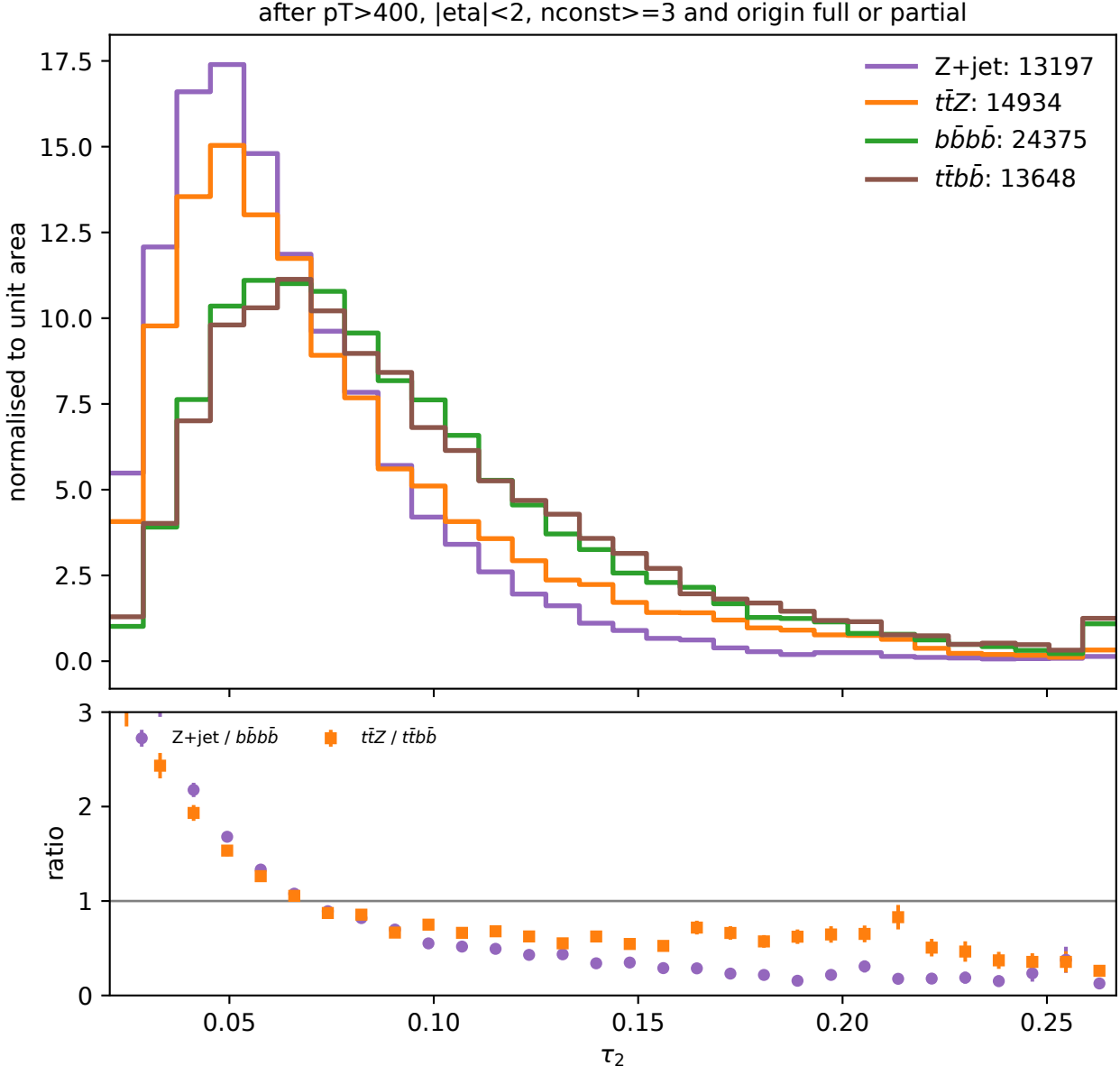

after pT>400, |eta|<2, nconst>=3 and origin full or partial
Z+jet: 13197
$t\bar{t}Z$: 14934
$b\bar{b}b\bar{b}$: 24375
$t\bar{t}b\bar{b}$: 13648
normalised to unit area
0.1
0.2
0.3
0.4
Z+jet / $b\bar{b}b\bar{b}$
$t\bar{t}Z$ / $t\bar{t}b\bar{b}$
ratio
0
1
2
3
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
jet η

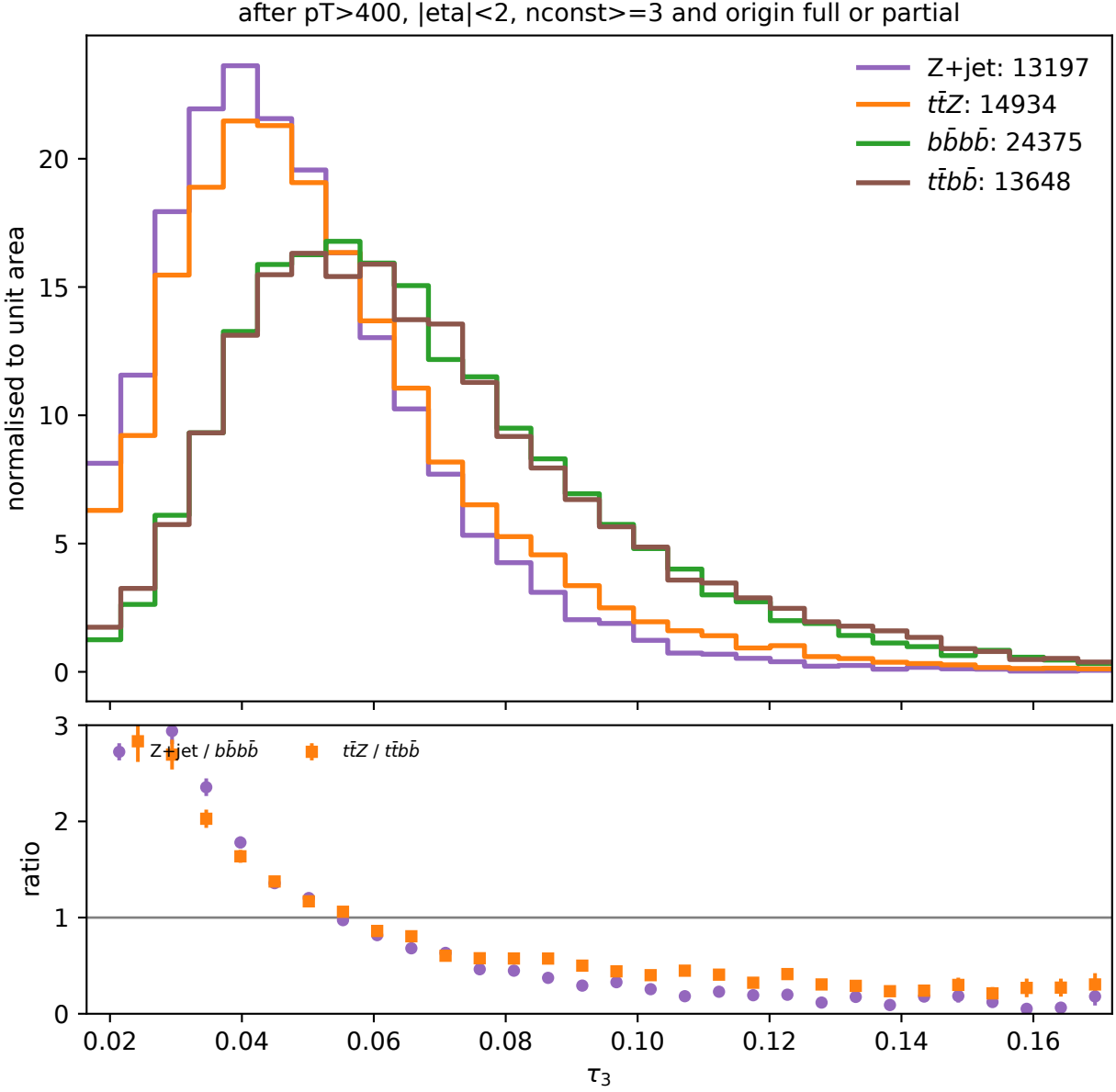

after pT>400, |eta|<2, nconst>=3 and origin full or partial
Z+jet: 13197
$t\bar{t}Z$: 14934
$b\bar{b}b\bar{b}$: 24375
$t\bar{t}b\bar{b}$: 14648
normalised to unit area
0.030
0.025
0.020
0.015
0.010
0.005
0.000
Z+jet / $b\bar{b}b\bar{b}$
$t\bar{t}Z$ / $t\bar{t}b\bar{b}$
ratio
3
2
1
0
50
100
150
200
250
jet mass [GeV]

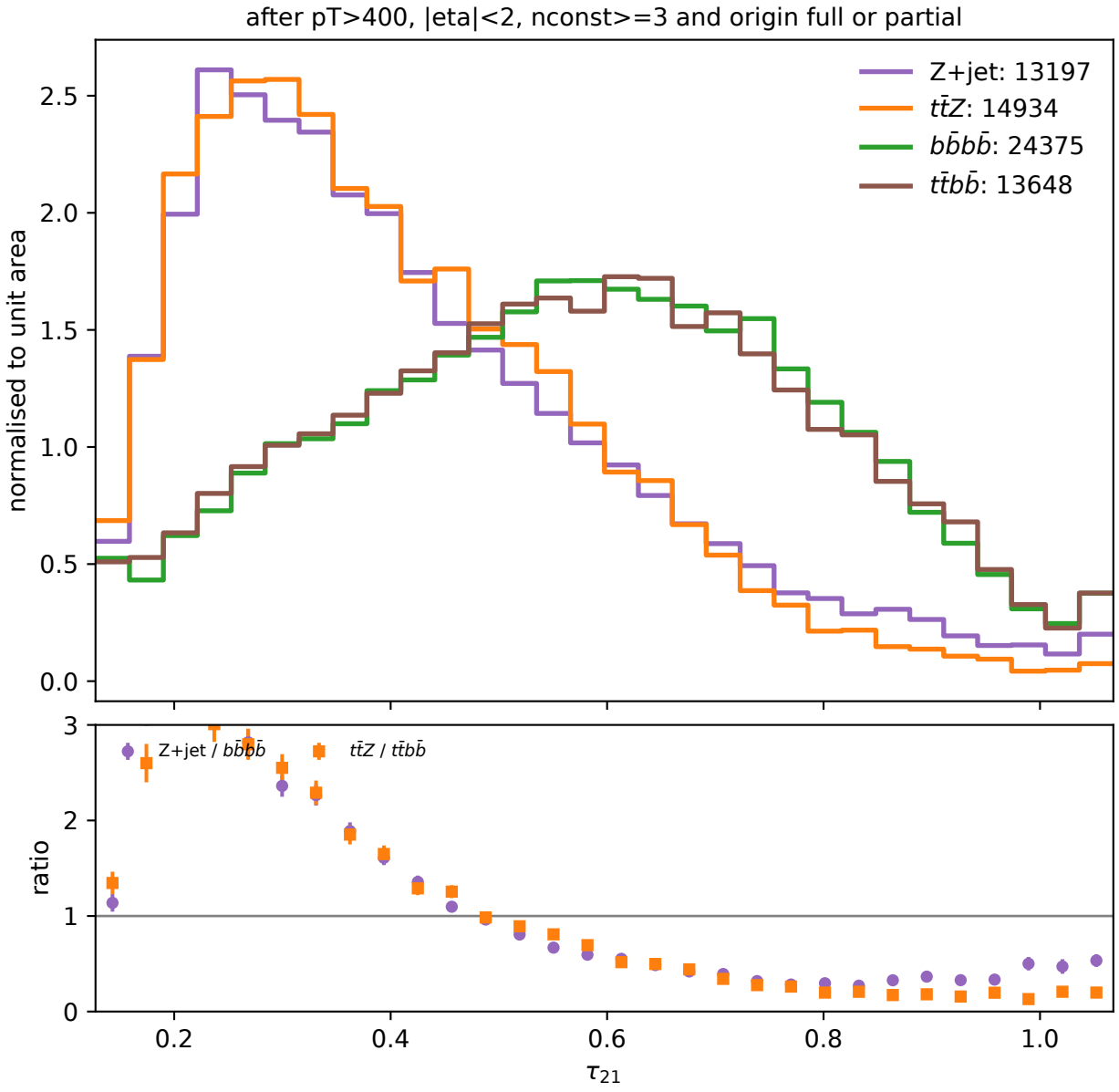

after pT>400, |eta|<2, nconst>=3 and origin full or partial
Z+jet: 13197
$t\bar{t}Z$: 14934
$b\bar{b}b\bar{b}$: 24375
$t\bar{t}b\bar{b}$: 14648
normalised to unit area
0.035
0.030
0.025
0.020
0.015
0.010
0.005
0.000
Z+jet / $b\bar{b}b\bar{b}$
$t\bar{t}Z$ / $t\bar{t}b\bar{b}$
ratio
3
2
1
0
20
30
40
50
60
70
80
90
100
constituent multiplicity

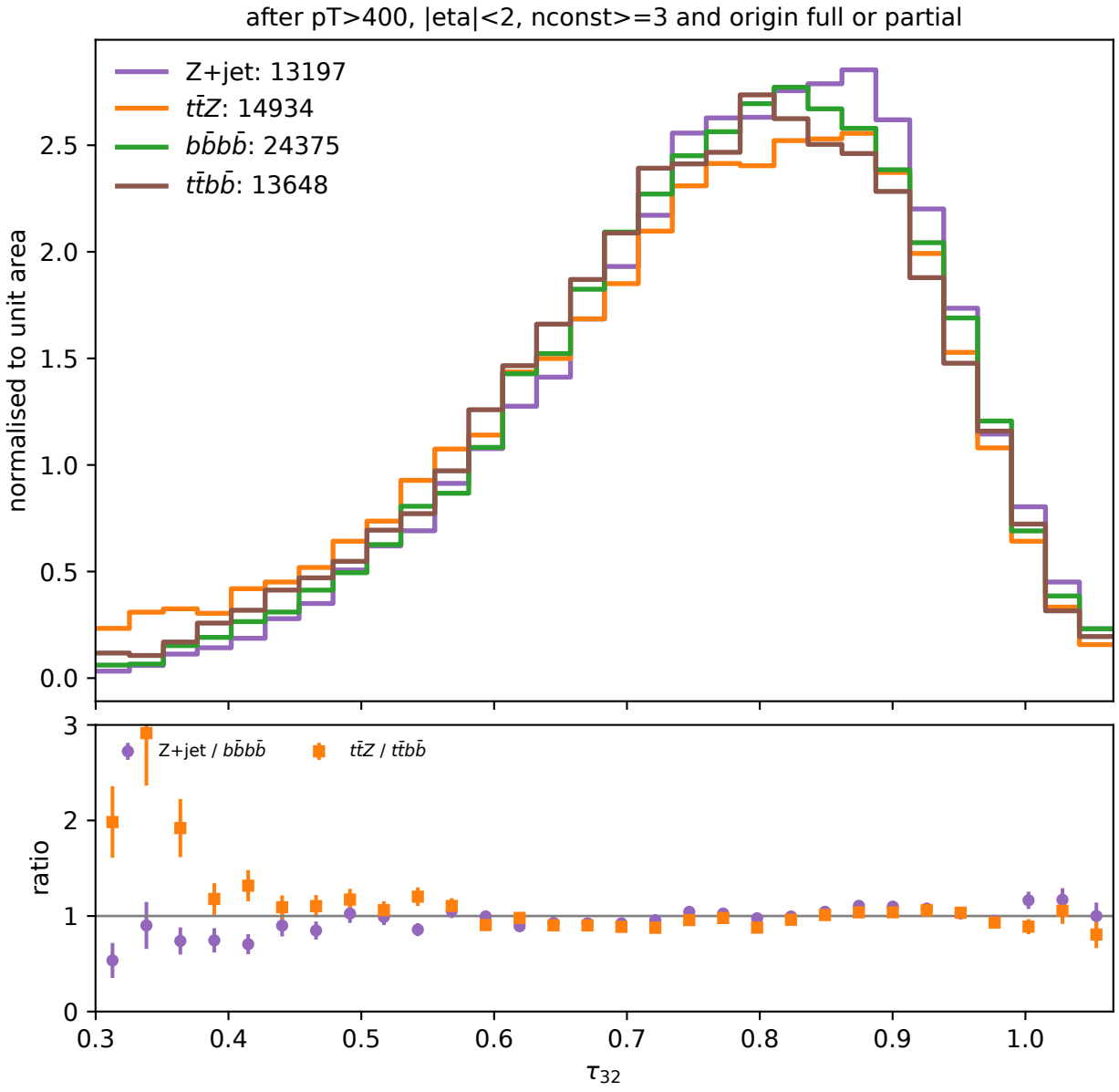

after pT>400, |eta|<2, nconst>=3 and origin full or partial
Z+jet: 13197
$t\bar{t}Z$: 14934
$b\bar{b}b\bar{b}$: 24375
$t\bar{t}b\bar{b}$: 13648
normalised to unit area
Z+jet / $b\bar{b}b\bar{b}$
$t\bar{t}Z$ / $t\bar{t}b\bar{b}$
ratio
girth

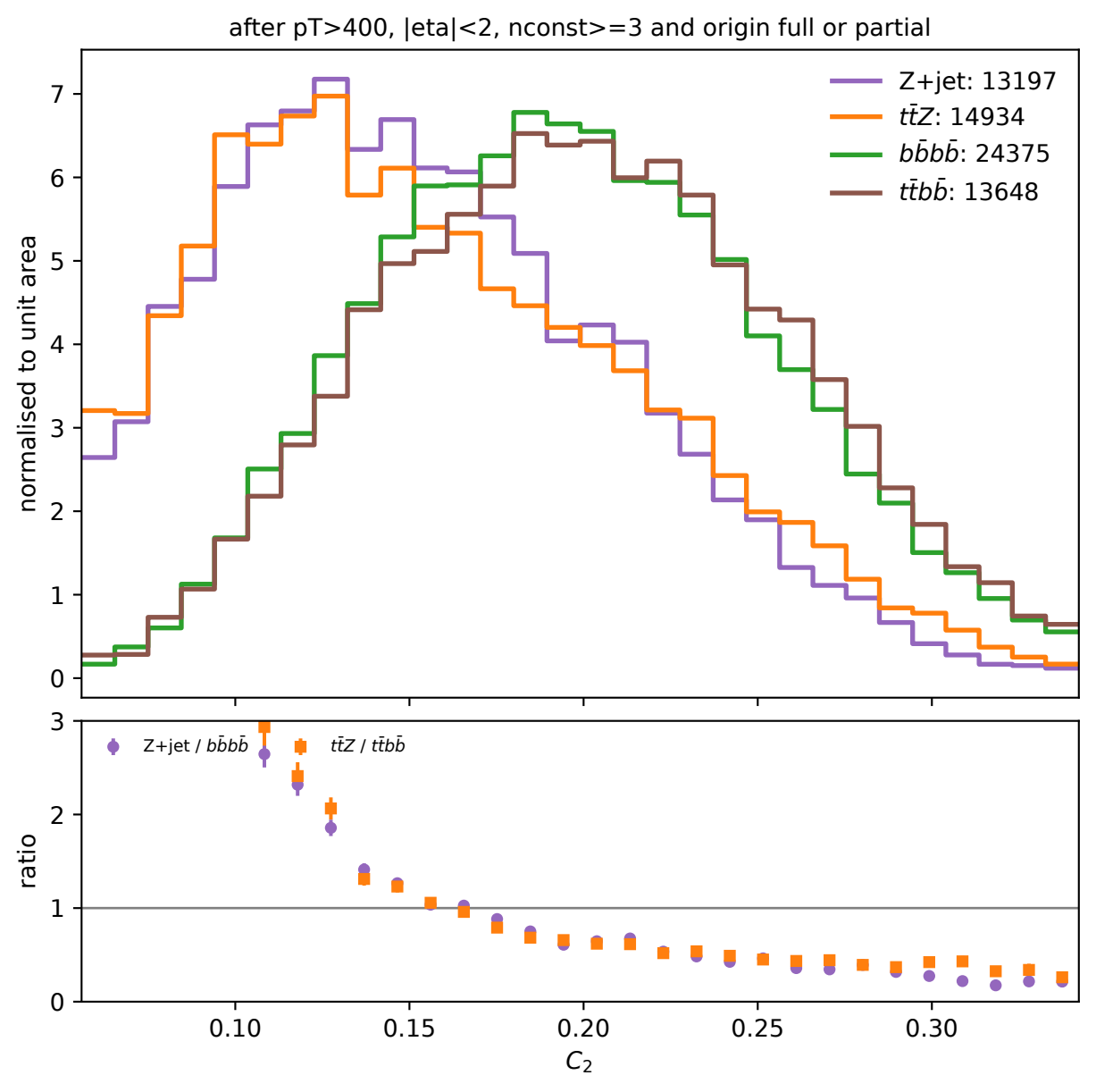

after pT>400, |eta|<2, nconst>=3 and origin full or partial
Z+jet: 13197
$t\bar{t}Z$: 14934
$b\bar{b}b\bar{b}$: 24375
$t\bar{t}b\bar{b}$: 13648
normalised to unit area
Z+jet / $b\bar{b}b\bar{b}$
$t\bar{t}Z$ / $t\bar{t}b\bar{b}$
ratio
τ_1

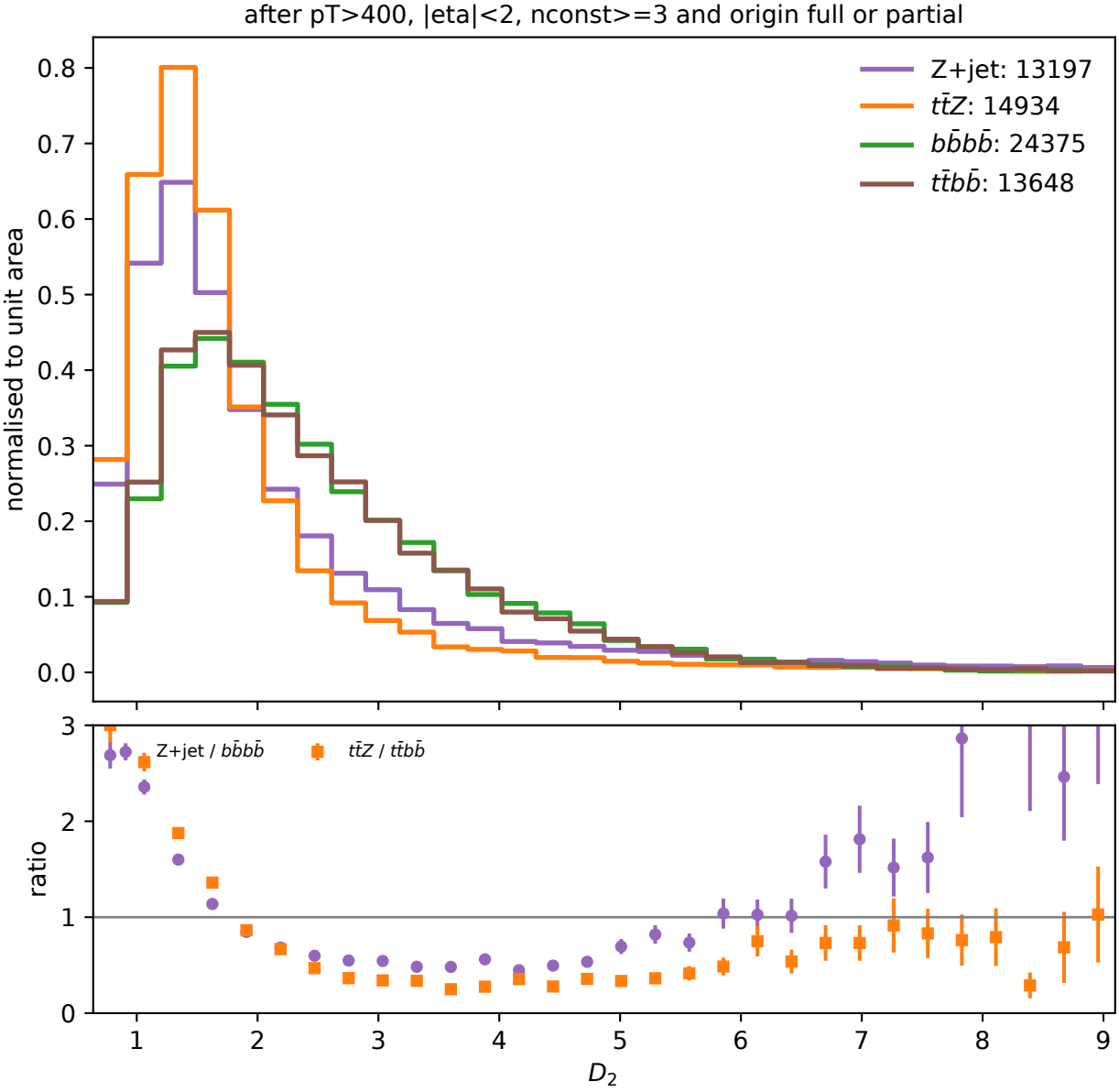

after pT>400, |eta|<2, nconst>=3 and origin full or partial
Z+jet: 13197
$t\bar{t}Z$: 14934
$b\bar{b}b\bar{b}$: 24375
$t\bar{t}b\bar{b}$: 14648
normalised to unit area
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
Z+jet / $b\bar{b}b\bar{b}$
$t\bar{t}Z$ / $t\bar{t}b\bar{b}$
ratio
3
2
1
0
0.05
0.10
0.15
0.20
0.25
τ_2

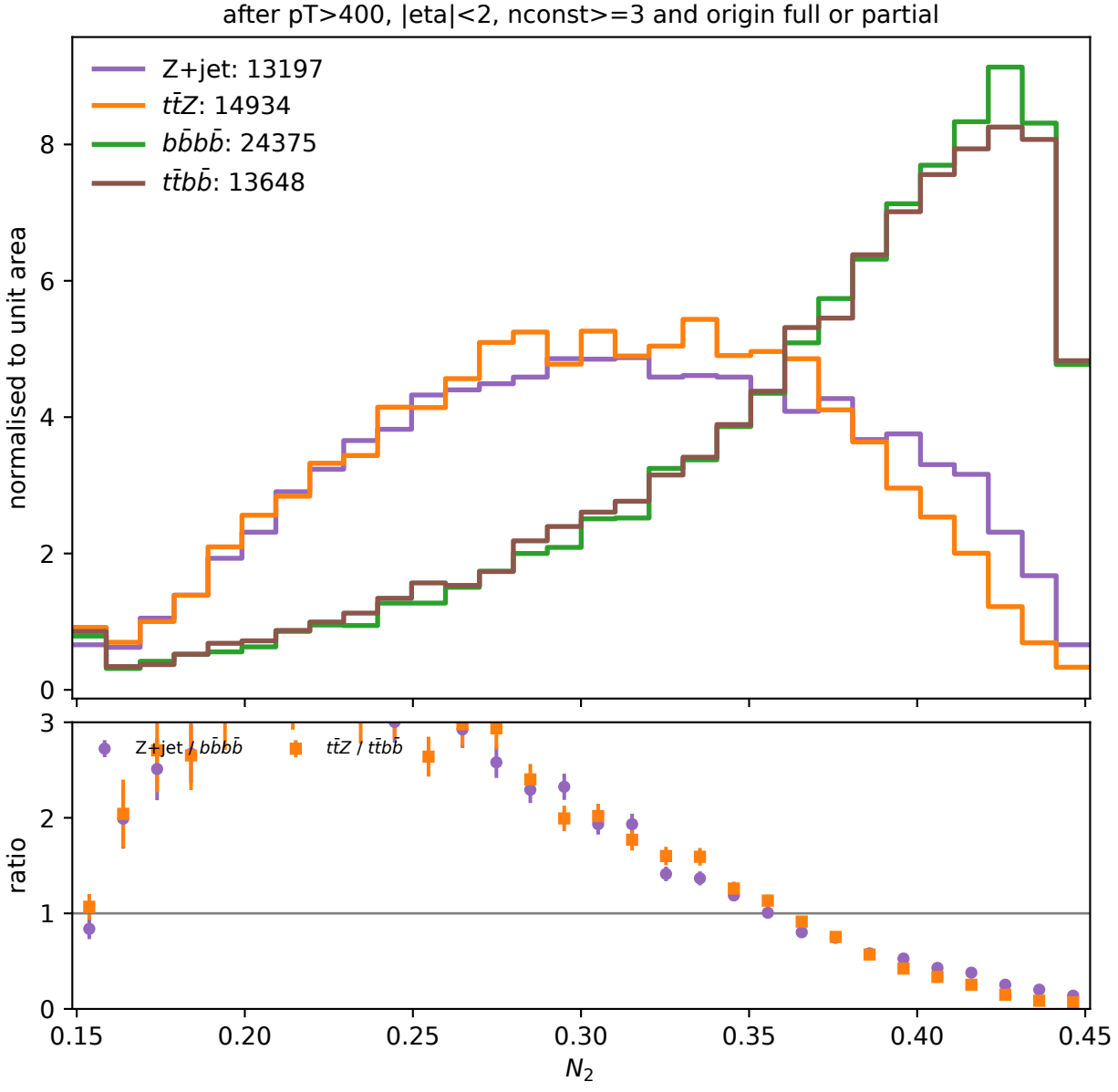

after pT>400, |eta|<2, nconst>=3 and origin full or partial
Z+jet: 13197
$t\bar{t}Z$: 14934
$b\bar{b}b\bar{b}$: 24375
$t\bar{t}b\bar{b}$: 14648
normalised to unit area
0
5
10
15
20
Z+jet / $b\bar{b}b\bar{b}$
$t\bar{t}Z$ / $t\bar{t}b\bar{b}$
ratio
0
1
2
3
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.16
τ_3

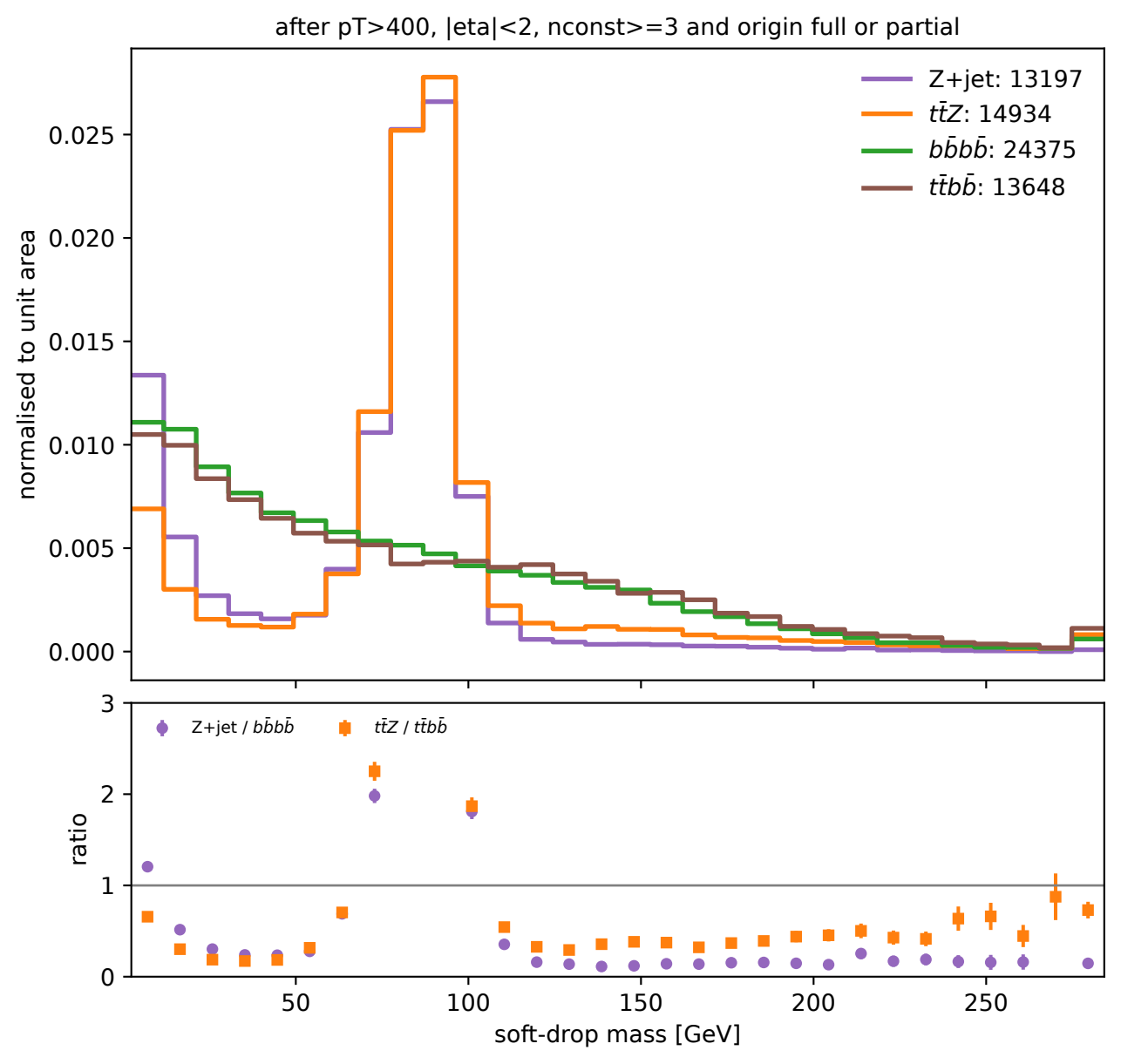

after pT>400, |eta|<2, nconst>=3 and origin full or partial
Z+jet: 13197
$t\bar{t}Z$: 14934
$b\bar{b}b\bar{b}$: 24375
$t\bar{t}b\bar{b}$: 14648
normalised to unit area
0.0
0.5
1.0
1.5
2.0
2.5
Z+jet / $b\bar{b}b\bar{b}$
$t\bar{t}Z$ / $t\bar{t}b\bar{b}$
ratio
0
1
2
3
0.2
0.4
0.6
0.8
1.0
τ_{21}

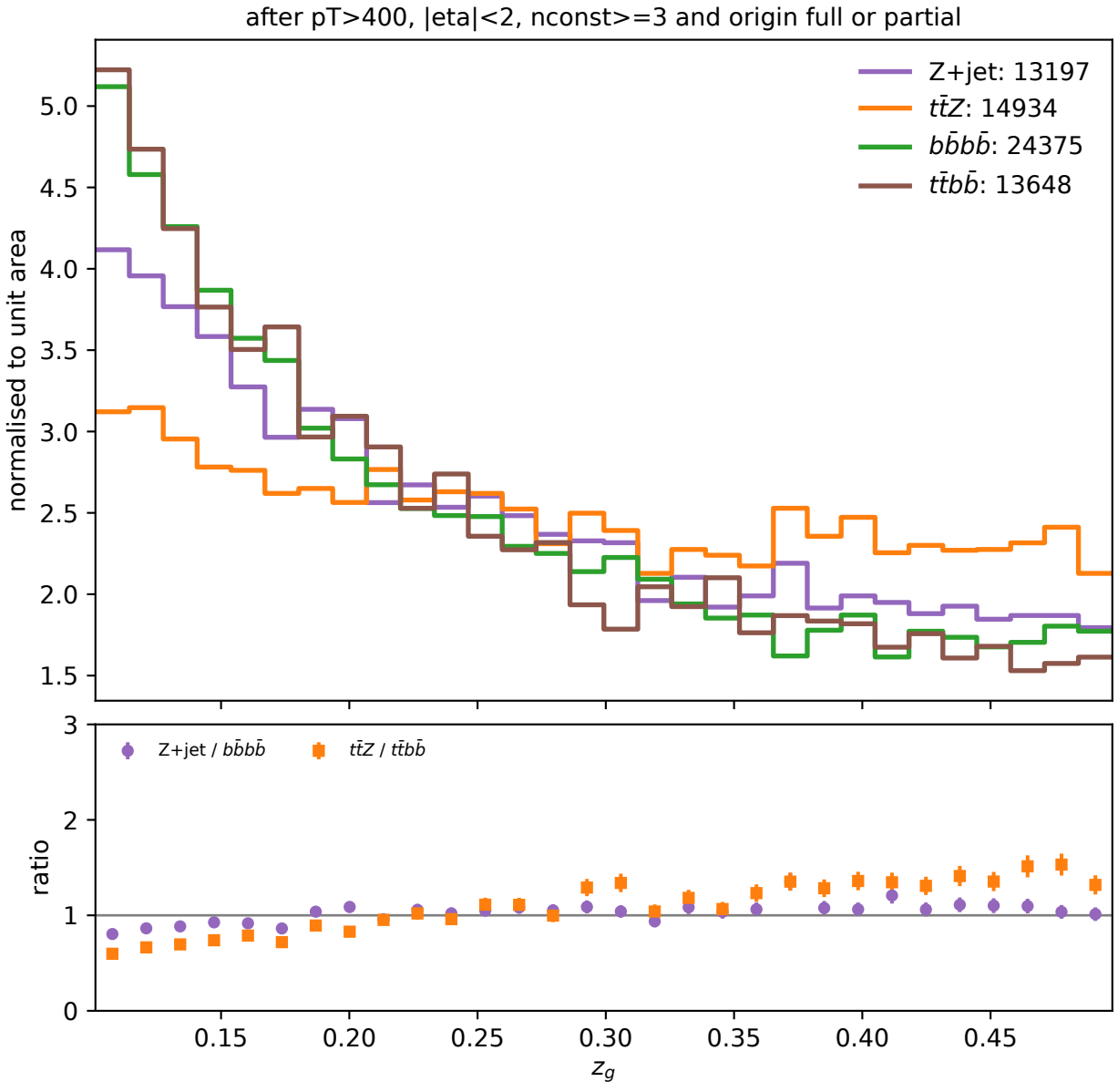

after pT>400, |eta|<2, nconst>=3 and origin full or partial
Z+jet: 13197
$t\bar{t}Z$: 14934
$b\bar{b}b\bar{b}$: 24375
$t\bar{t}b\bar{b}$: 14648
normalised to unit area
0.0
0.5
1.0
1.5
2.0
2.5
Z+jet / $b\bar{b}b\bar{b}$
$t\bar{t}Z$ / $t\bar{t}b\bar{b}$
ratio
0
1
2
3
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
τ_{32}

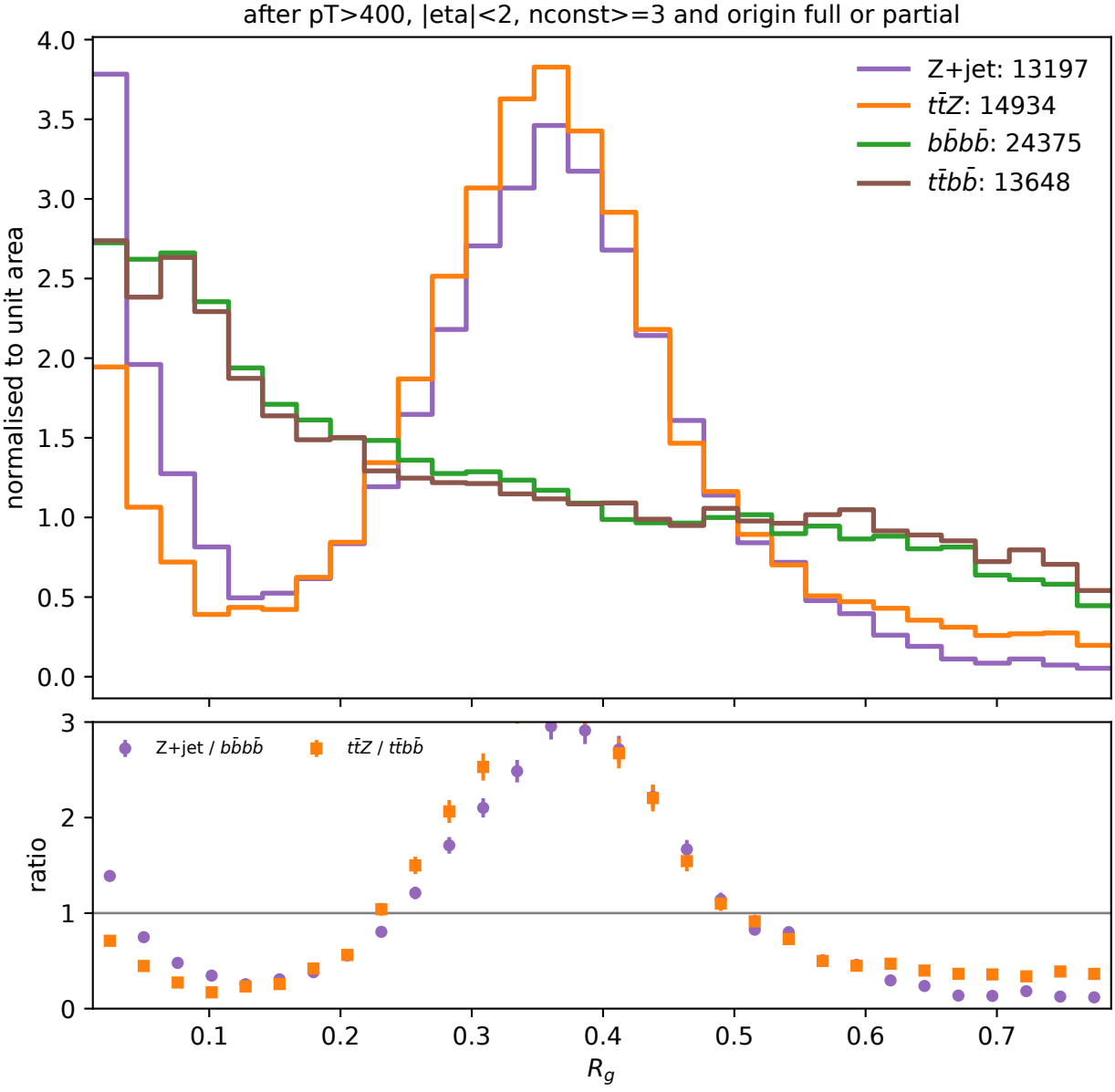

after pT>400, |eta|<2, nconst>=3 and origin full or partial
Z+jet: 13197
$t\bar{t}Z$: 14934
$b\bar{b}b\bar{b}$: 24375
$t\bar{t}b\bar{b}$: 14648
normalised to unit area
Z+jet / $b\bar{b}b\bar{b}$
$t\bar{t}Z$ / $t\bar{t}b\bar{b}$
ratio
C_2


after pT>400, |eta|<2, nconst>=3 and origin full or partial
Z+jet: 13197
$t\bar{t}Z$: 14934
$b\bar{b}b\bar{b}$: 24375
$t\bar{t}b\bar{b}$: 14648
normalised to unit area
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Z+jet / $b\bar{b}b\bar{b}$
$t\bar{t}Z$ / $t\bar{t}b\bar{b}$
ratio
0
1
2
3
1
2
3
4
5
6
7
8
9
D_2


after pT>400, |eta|<2, nconst>=3 and origin full or partial
Z+jet: 13197
$t\bar{t}Z$: 14934
$b\bar{b}b\bar{b}$: 24375
$t\bar{t}b\bar{b}$: 14648
normalised to unit area
0
2
4
6
8
Z+jet / $b\bar{b}b\bar{b}$
$t\bar{t}Z$ / $t\bar{t}b\bar{b}$
ratio
0
1
2
3
0.15
0.20
0.25
0.30
0.35
0.40
0.45
N_2


after pT>400, |eta|<2, nconst>=3 and origin full or partial
Z+jet: 13197
$t\bar{t}Z$: 14934
$b\bar{b}b\bar{b}$: 24375
$t\bar{t}b\bar{b}$: 14648
normalised to unit area
0.025
0.020
0.015
0.010
0.005
0.000
Z+jet / $b\bar{b}b\bar{b}$
$t\bar{t}Z$ / $t\bar{t}b\bar{b}$
ratio
3
2
1
0
50
100
150
200
250
soft-drop mass [GeV]


after pT>400, |eta|<2, nconst>=3 and origin full or partial
Z+jet: 13197
$t\bar{t}Z$: 14934
$b\bar{b}b\bar{b}$: 24375
$t\bar{t}b\bar{b}$: 14648
normalised to unit area
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
Z+jet / $b\bar{b}b\bar{b}$
$t\bar{t}Z$ / $t\bar{t}b\bar{b}$
ratio
3
2
1
0
0.15
0.20
0.25
0.30
0.35
0.40
0.45
z_g


after pT>400, |eta|<2, nconst>=3 and origin full or partial
Z+jet: 13197
$t\bar{t}Z$: 14934
$b\bar{b}b\bar{b}$: 24375
$t\bar{t}b\bar{b}$: 13648
normalised to unit area
Z+jet / $b\bar{b}b\bar{b}$
$t\bar{t}Z$ / $t\bar{t}b\bar{b}$
ratio
R_g